

Xiaodong Liu

+61 472 737 067 | lxd040715@gmail.com | LinkedIn | GitHub | Portfolio
Melbourne, Australia

PROFESSIONAL SUMMARY

Full-stack software engineer specializing in building **AI-driven** modern web and mobile applications. Proficient in **Next.js 15**, **FastAPI**, **Node.js**, and various databases (**PostgreSQL**, **Qdrant**), with end-to-end experience from architectural design to deployment. Expert in implementing **RAG (Retrieval-Augmented Generation)** in real-world scenarios such as e-commerce assistants and knowledge management. Leverages AI-assisted development tools to significantly boost engineering efficiency. Possesses strong cross-platform development skills (Web/Android) and a business-oriented mindset to solve complex technical challenges.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Kotlin, Python, Java, C
- **AI & Engineering:** **RAG System Design**, **LLM API Integration** (Claude, OpenAI), **Prompt Engineering**, **Vector Search (Qdrant)**, AI-Assisted Development (Codex, TRAE, Cursor)
- **Backend & APIs:** Node.js, **FastAPI**, **Next.js (App Router)**, **REST APIs**, **SSE (Streaming)**, **JWT**, **OAuth 2.0**
- **Frontend & Mobile:** **React 19**, **Jetpack Compose (Android Native)**, **Tailwind CSS**, **Bootstrap**, **HTML5/CSS3**
- **Database & Cloud:** **PostgreSQL**, **MongoDB**, **Prisma ORM**, **Mongoose**, **Upstash Redis**, **Vercel**, **Render**, **AWS**, **Cloudflare R2**, **Cloudinary**

EXPERIENCE & PROJECTS

ShopMate — RAG-based AI Conversational E-commerce Assistant (Android) 2026

Full-Stack Lead Developer

Tech Stack: Android Native, Kotlin, Jetpack Compose, Node.js, TypeScript, Express, PostgreSQL, Qdrant, Claude/OpenAI LLM, SSE

- **Core Architecture Design:** Designed and implemented the end-to-end architecture as lead developer, covering Android native client, Node.js backend, RAG retrieval, and AI integration modules to ensure system stability under high concurrency.
- **RAG Pipeline Optimization:** Integrated traditional e-commerce workflows into a conversational AI assistant. Leveraged **Qdrant vector search** and **PostgreSQL fact-checking** to eliminate LLM hallucinations, achieving an ultra-fast **first-token response within 1s**.
- **End-to-End E-commerce Lifecycle:** Built a complete shopping journey from product recommendation and details to cart management and simulated checkout. Utilized **Jetpack Compose** for a fluid UI and **SSE** for real-time streaming of AI responses and business states.
- **Engineering Evaluation:** Established a **RAG Baseline evaluation system**. Optimized query rewriting and filtering logic through trace tracking and Gradio workbenches, significantly improving recall and explainability of recommendations.

DevStash — Full-Stack Knowledge Management SaaS for Developers 2026

Independent Developer

Tech Stack: Next.js 15, React 19, TypeScript, Prisma 7, Neon PostgreSQL, Auth.js, Stripe, Cloudflare R2, Upstash Redis

- **SaaS Platform Development:** Developed a full-stack knowledge management platform supporting structured storage for code snippets, prompts, and multimedia assets. Implemented Role-Based Access Control (**RBAC**) and **Stripe subscription billing**.
- **AI Feature Integration:** Integrated LLM APIs for automated tag generation, intelligent content summarization, and code logic explanation. Implemented rate limiting via **Upstash Redis** to ensure system stability during traffic spikes.
- **Deployment & Automation:** Managed data modeling, automated testing (**Vitest**), and **Vercel** production deployment. Optimized CI/CD pipelines to accelerate engineering iteration cycles.

PropertyPulse — Full-Stack Real Estate Listing Management Platform 2025

Full-Stack Developer

Tech Stack: Next.js, React, MongoDB, Mongoose, NextAuth, Cloudinary, MapTiler

- **Listing Management System:** Developed a responsive real estate management system integrating **Google OAuth** authentication, **Cloudinary** multimedia management, and **MapTiler** interactive maps to achieve a full business loop from listing to search.

EDUCATION

University of Melbourne

2023.03 – 2026.11

Bachelor of Science, Major in Computer Science

Melbourne, Australia

- **Core Coursework:** Algorithms and Data Structures, Software Modeling, Database Systems, Foundations of Artificial Intelligence, Network Computing, etc.